

A Graph Model for Sorting Signed Permutation by Reversal Problem *

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Abstract

Computing reversal distance of two signed permutations has given rise to computational molecular biology with regard to genome rearrangement in the recent decade. Given two genomes represented as signed permutations of the same elements, the problem consists in finding the minimum number of reversals that transforms one genome into the other. Based on this problem, we propose a Signed Reversal graph to discuss some related problems with sorting signed permutations by reversals in graph theoretical views. Graph properties including connectivity, diameter and fault diameter are investigated in this paper.

Keywords: Computational biology, Genome rearrangement, Signed permutations, Sorting by reversals, Signed Reversal graphs, Graph theory.

1 Introduction

The problem of sorting signed permutation by reversals is inspired and motivated by comparative genomics. Given two genomes represented as signed permutations of the same elements, the problem consists in finding the minimum number of reversals that transforms one genome into the other. Biologists use methods that solve this problem to estimate an evolutionary distance between genomes.

Given a genome with n genes, the order of genes can be represented by a permutation $\pi = \pi_1\pi_2 \dots \pi_n$. In the case of *signed* permutations, each element in π is associated with +

or - signs, or none for + sign. In this paper, for convenience, we define $\overline{\pi_i}$ as $-\pi_i$. Every signed reversal results not only in reversing of any number of consecutive ordered terms, but also changing the sign of each term within the scope of the reversal. For the permutation of n numbers, let $\rho(i, j) = 12 \dots i - 1 \overline{j} j - 1 \dots i + 1 \overline{i} i + 1 \dots n$. Then $\pi \cdot \rho(i, j)$ is the permutation obtained from π by reversing the order and flipping the signs of the elements in the interval $[i, j]$, i.e., in effect, a reversal $\rho(i, j)$ transforms π into $\pi \cdot \rho(i, j)$.

$$\pi = \pi_1 \dots \pi_{i-1} \pi_i \pi_{i+1} \dots \pi_{j-1} \pi_j \pi_{j+1} \dots \pi_n$$

$$\pi \cdot \rho(i, j) = (\pi_1 \pi_2 \dots \pi_{i-1} \overline{\pi_j} \overline{\pi_{j-1}} \dots \overline{\pi_{i+1}} \overline{\pi_i} \pi_{j+1} \dots \pi_n).$$

Sorting a signed permutation π by reversals problem has been a popular issue in the computational biology in literatures. We notice that many researchers used the graph structure [2, 7, 9] to analysis the sorting signed permutation by reversals problem. In this paper, we propose a new graph model as a tool to figure out the problem or phenomenon about the sorting signed permutation by reversals problem.

This paper is organized as follows. In the next section, the necessary terminology in the graph theory and the formal definition of Signed Reversal Graph are introduced. In section 3, we analysis the structure of Signed Reversal Graph and discuss some of its properties. The connectivity and diameter of Signed Reversal Graph is proved in section 4. Finally, we give a conclusion.

2 Preliminary

For the graph definition and notation we follow [4, 8]. $G = (V, E)$ is a simple graph if V is a finite set and E is a subset of $\{(u, v) | (u, v) \text{ is}$

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an unordered pair of V }. V is the *vertex set* and E is the *edge set*. The *neighborhood* of u , $N_G(u)$, is $\{v \mid (u, v) \in E\}$. The *degree* of a vertex of a graph is the number of edges incident to the vertex. A *regular graph* is a graph where each vertex has the same number of edges incident to it, i.e., every vertex has the same degree. A regular graph with vertices of degree k is denoted as k -regular graph. The complete graph is a simple graph in which every pair of distinct vertices is connected by an edge, and it is denoted by K_n if it has n vertices. Two vertices u and v are *adjacent* if $(u, v) \in E$. A *path* is a sequence of adjacent vertices, written as $\langle v_0, v_1, v_2, \dots, v_m \rangle$, in which all the vertices $v_0, v_1, v_2, \dots, v_m$ are distinct except possibly $v_0 = v_m$. For this path, v_0 and v_m are called *end vertices*, and $v_i, 1 \leq i \leq m-1$, is called *internal vertex*. We also write a path with the end vertices u, v as (u, v) -path. Suppose that n (u, v) -paths $P_1, P_2, \dots, P_n$ have no common internal vertices, they are said to be *internally disjoint*, or *disjoint* if no ambiguous. The *length* of a path P is the number of edges in P . The *distance* between two vertices u, v in a graph G , $d_G(u, v)$, is the number of edges in a shortest path connecting them. The *diameter* of a graph G , $D(G)$, is the maximum distance between any two vertices.

In an undirected graph G , two vertices u and v are called *connected* if G contains a path from u to v . Otherwise, they are called *disconnected*. A graph is called *connected* if every pair of distinct vertices in the graph can be connected through some path. A subset C of V is called a *vertex-cut* if $G - C$ is either disconnected or an isolated vertex. It is known that only complete graphs do not have vertex cuts. The *connectivity* of G , written $\kappa(G)$, is defined as the minimum cardinality over all vertex-cuts. A graph G is n -connected if $\kappa(G) = n$. For the property of connectivity, Menger proved the following theorem.

Theorem 1 (Menger's Theorem) Let $G = (V, E)$ is a graph. The connectivity $\kappa(G) = n$ if and only if there exists n internally disjoint (u, v) -paths over all pair of vertices $u, v \in V(G)$.

Assume that G is a k -regular graph with connectivity κ . We say that G is *maximum-connected* if $\kappa = k$; and G is *super-connected* if it is a complete graph, or it is maximum connected and every minimum vertex cut is $N_G(v)$ for some vertex v . A *perfect matching* between two graph G_1 and G_2 is a one-to-one and onto matching on $V(G_1)$ and $V(G_2)$. That is, every vertex of the graphs G_1 and G_2 is incident to exactly one edge of the

matching. Let M be any perfect matching between G_1 and G_2 . Then $G(G_1, G_2; M)$ is a graph constructed with G_1 and G_2 by connecting each vertex of G_1 to the vertex of G_2 via the matching M . That is $V(G(G_1, G_2; M)) = V(G_1) \cup V(G_2)$ and $E(G(G_1, G_2; M)) = E(G_1) \cup E(G_2) \cup M$. For perfect matching, Chen et. al. proved the following property as Theorem 2.

Theorem 2 [6] Let G_1 and G_2 be two k -regular maximum-connected graphs with t vertices, and M be any perfect matching between $V(G_1)$ and $V(G_2)$. Then, $G(G_1, G_2; M)$ is $(k+1)$ -regular super-connected if and only if (1) $t > k+1$ or (2) $t = k+1$ with $k = 0, 1, 2$.

Then, we give the formal definition of the Signed Reversal Graphs as follows.

Definition 1 A n -dimensional Signed Reversal Graph, denoted by SR_n , is a graph $G = (V, E)$. The vertex set $V = \{\pi_1 \pi_2 \dots \pi_n \mid \pi_i \in Z_n \text{ and } |\pi_i| \neq |\pi_j| \text{ for } i \neq j\}$ and the edge set $E = \{(u, v) \mid v = u \cdot \rho(i, j), \text{ and } 1 \leq i \leq j \leq n\}$.

Figure 1 and 2 illustrate the SR_1, SR_2 , and SR_3 , respectively.

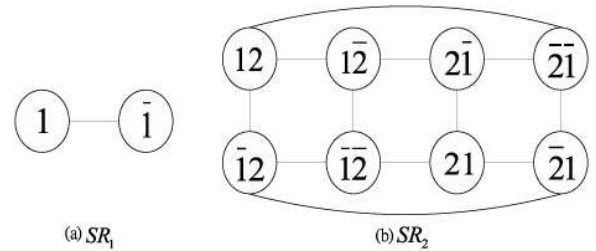


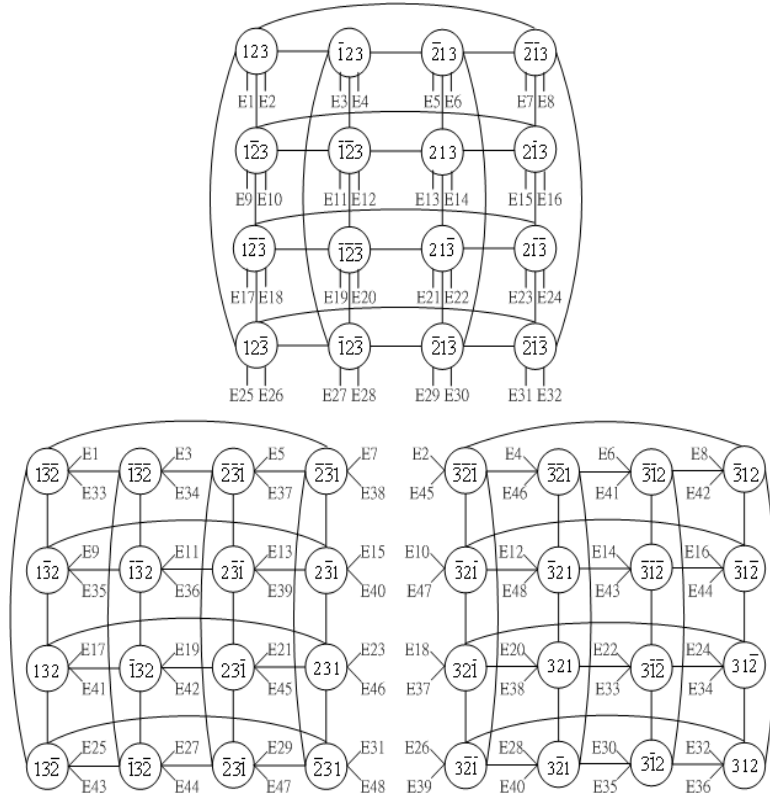
Figure 1: Illustration of SR_1 and SR_2 .

Property 1 SR_n has $2^n \cdot n!$ vertices.

Property 2 SR_n is a $\frac{n(n+1)}{2}$ -regular graph.

3 Topological properties

Let $V(SR_{n-1}^i), 1 \leq |i| \leq n$, be the subset of $V(SR_n)$ induced by the vertices which the left-most label of the vertices is i , i.e., $V(SR_{n-1}^i) = \{i\pi_2 \dots \pi_n \mid \pi_j \in Z_n - \{i, \bar{i}\} \text{ and } |\pi_j| \neq i \text{ for } 2 \leq j \leq n\}$. For example, if $n = 2$ and $i = 1$, then $V(SR_2) = \{12, 1\bar{2}, \bar{1}2, \bar{1}\bar{2}, 21, 2\bar{1}, \bar{2}1, \bar{2}\bar{1}\}$ and $V(SR_1^1) = \{12, 1\bar{2}\}$. Note that $V(SR_n) =$

Figure 2: Illustration of SR_3 .

$V(SR_{n-1}^1) \cup \dots \cup V(SR_{n-1}^n) \cup V(SR_{n-1}^{\bar{1}}) \cup \dots \cup V(SR_{n-1}^{\bar{n}})$. Now, considering $u = \pi_1 \pi_2 \dots \pi_{n-1} \pi_n \in V(SR_n)$, then $v = u \cdot \rho(1, k) \in V(SR_{n-1}^{\bar{k}})$ for $k = 1, 2, \dots, n-1, n$. Therefore, the recursive definition about n -dimensional signed reversal graph is given as follows.

Definition 2 Let n be a positive integer. The n -dimensional Signed Reversal Graph SR_n has $2^n \cdot n!$ vertices, each is labeled by a n -bit number, $\{\pi_1 \pi_2 \dots \pi_n \mid \pi_i \in Z_n \text{ and } |\pi_i| \neq |\pi_j| \text{ for } i \neq j\}$. We define $SR_1 = K_2$. For $n \geq 1$, SR_{n+1} is obtained by taking $2(n+1)$ copies of the signed reversal graph SR_n , denoted by $SR_n^1, SR_n^2, \dots, SR_n^{n+1}, SR_n^{\bar{1}}, SR_n^{\bar{2}}, \dots, SR_n^{\bar{n+1}}$, adding $2^{n-1} \cdot n!$ edges between SR_n^p and SR_n^q for $1 \leq |p| \neq |q| \leq n+1$, and adding $2^n \cdot n!$ edges between SR_n^p and SR_n^q for $p = \bar{q}$ by the following rules. A vertex $u = u_1 u_2 \dots u_{n+1}$ of SR_n^p is jointed to a vertex v of SR_n^q iff $v = u \cdot \rho(1, k)$ for $\bar{u}_k = q$.

For example, 123 is jointed to $\bar{1}23$ of $SR_2^{\bar{1}}$, $\bar{2}13$ of $SR_2^{\bar{2}}$, and $\bar{3}2\bar{1}$ of $SR_2^{\bar{3}}$.

By the above definition, SR_{n+1} can be viewed as the union of the $2(n+1)$

subgraphs SR_n , and we write this recursive construction symbolically as $SR_{n+1} = SR_n^1 \diamond SR_n^2 \diamond \dots \diamond SR_n^{n+1} \diamond SR_n^{\bar{1}} \diamond SR_n^{\bar{2}} \diamond \dots \diamond SR_n^{\bar{n+1}}$.

Then we are going to discuss some properties of SR_n . First we show that SR_n is vertex symmetric by proving that SR_n is Cayley graph[5]. Let Γ be a finite group and let $S \subseteq \Gamma$ be a set of generators of Γ not containing the identity and closed under inversion, that is if $s \in S$, then $s^{-1} \in S$ also. Then the graph $Cay(\Gamma, S)$ is a Cayley graph. Let $\Gamma = \{1, 2, \dots, n\}$ and $\bar{\Gamma} = \{\bar{1}, \bar{2}, \dots, \bar{n}\}$. Let $\Gamma^R = \Gamma \cup \bar{\Gamma}$ and $\Gamma_n^R = \{\alpha = \alpha_1 \alpha_2 \dots \alpha_n \mid \alpha_i \in \Gamma^R \text{ and } |\alpha_i| \neq |\alpha_j| \text{ for } 1 \leq i \neq j \leq n\}$. Let the generator be $\rho(i, j) = 12 \dots i-1 \bar{j} j-1 \dots i+1 \bar{i} j+1 \dots n$ for $1 \leq i, j \leq n$. Clearly, $\rho(i, j) \in \Gamma_n^R$. Then the set of generators is $S_n = \{\rho(i, j) \mid 1 \leq i, j \leq n\}$. Thus, the identity of Γ_n^R is $12 \dots n$, $\alpha^{-1} = \alpha$, and $\alpha \cdot \rho(i, j) \in \Gamma_n^R$. Then, we can define that $SR_n = Cay(\Gamma_n^R, S_n)$. Therefore, SR_n is a Cayley graph. Since Cayley graph is vertex symmetric, SR_n is vertex symmetric.

Lemma 1 SR_n is vertex symmetric.

By the recursive definition, we have the following observation.

Property 3 *There exists a perfect matching between each pair of SR_n^i and $SR_n^{\bar{i}}$ for $i = 1, 2, \dots, n+1$.*

Property 4 *Let $u, v \in V(SR_n^i)$, $u', v' \in V(SR_n^{\bar{i}})$, and $(u, u'), (v, v') \in E(SR_{n+1})$. Suppose that $(u, v) \in E(SR_n^i)$. Then $(u', v') \in E(SR_n^{\bar{i}})$.*

With Property 3, we can divide SR_{n+1} into $(n+1)$ groups as follows. Let $g_i = (SR_n^i, SR_n^{\bar{i}})$, i.e., $g_1 = (SR_n^1, SR_n^{\bar{1}})$, $g_2 = (SR_n^2, SR_n^{\bar{2}})$, $\dots$, $g_{n+1} = (SR_n^{n+1}, SR_n^{\bar{n+1}})$. The group structure of SR_2, SR_3, SR_4 and SR_{n+1} is illustrated in figures 3-5.

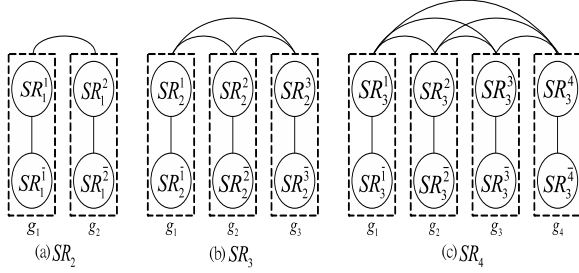


Figure 3: Illustration of the structure of SR_2, SR_3 and SR_4 .

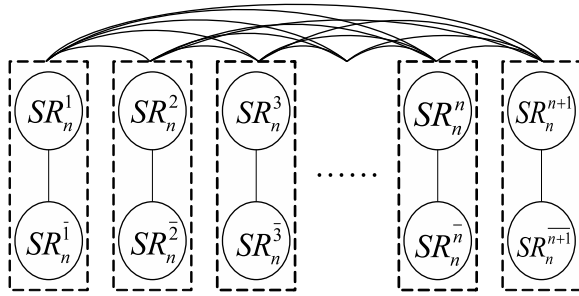


Figure 4: Illustration of the structure of SR_{n+1} .

By group view point of SR_{n+1} , we have the following property.

Property 5 *There exists a perfect matching between each pair of group g_i and g_j for $1 \leq i \neq j \leq n+1$ in SR_{n+1} .*

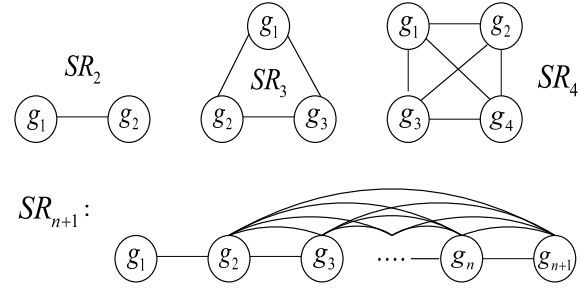


Figure 5: Illustration of the simple structure of SR_2, SR_3, SR_4 , and SR_{n+1} .

4 Connectivity and Diameter

In this section, we prove that the connectivity of SR_n is $\frac{n(n+1)}{2}$ and give a bound of some diameter properties. First, we need the following terms. For two distinct vertices u, v of a graph G , the *fault distance*, $d^f(u, v)$, between u and v is the maximum distance resulting from the deletion of any set of vertices, except u, v , containing less than $\kappa(G)$ vertices. The *fault diameter* of a graph G , $D^f(G)$, is the maximum fault distance between any two vertices of G . For two distinct vertices u, v of G , the *k-wide distance*, $\omega_k(u, v)$, between u and v is the minimum value l such that there exist k internally disjoint (u, v) -paths whose lengths are at most l . The *k-wide diameter*, $\omega_k(G)$, of G is the maximum value of the k -wide distances between any two vertices of G .

Since some reversal results in a genome rearrangement problem are impossible, i.e. the important function of some genes are disappear, it is reasonable to discuss the fault reversal distance in the sorting by reversals problem. The *k-fault reversal distance*, $d_k^f(\pi)$, of a permutation π is the maximum reversal distance resulting from the k reversal results that are not allow to appear. In SR_n , the fault distance between two vertices is the same concept. Thus, the fault diameter of SR_n is an interesting problem in genome rearrangement problem.

Theorem 3 *The connectivity of SR_n is $\frac{n(n+1)}{2}$.*

Proof: By Megner's Theorem, proving that there are $\frac{n(n+1)}{2}$ internally disjoint (u, v) -paths over all pair of vertices $u, v \in V(SR_n)$ is equivalent to prove that the connectivity of SR_n is $\frac{n(n+1)}{2}$. We prove this theorem by induction on n . For $n = 1$, there is only a path of SR_1 . For $n = 2$, since

SR_2 is a 3-dimensional hypercube[10], the connectivity of SR_2 is $3 = \frac{2(2+1)}{2}$ and SR_2 is also super-connected. Therefore, the base case is true. Assume that the theorem is true for $n = k$, i.e., there are $\frac{k(k+1)}{2}$ internally disjoint (u, v) -paths of SR_k . Then for $n = k + 1$, we divide the prove into two cases by the positions between any two vertices u and v in SR_n .

Case 1: u and v are in the same group, say g_i .

Since SR_k^i and $SR_k^{\bar{i}}$ are two $\frac{k(k+1)}{2}$ -regular maximum connected graphs with $2^k \cdot k! > \frac{k(k+1)}{2} + 1$ vertices for $k \geq 2$ and there exists a perfect matching M_1 between $V(SR_k^i)$ and $V(SR_k^{\bar{i}})$, by Theorem 2, g_i is $[\frac{k(k+1)}{2} + 1]$ -regular and super-connected, i.e., there are $\frac{k(k+1)}{2} + 1$ internally disjoint paths between u and v of g_i . Let u^{g_j} be the neighbor of u in g_j and v^{g_j} be the neighbor of v in g_j for $g_j \neq g_i$. By property 5, every group g_j have exactly one vertex u^{g_j} and one vertex v^{g_j} connected to u and v respectively for $g_j \neq g_i$. Clearly, there exists a path P^{g_j} joining u^{g_j} and v^{g_j} in g_j . Then for k group, there are k disjoint paths joining u and v in $SR_n - g_i$. Since g_i have $\frac{k(k+1)}{2} + 1$ disjoint (u, v) -paths, there are $\frac{k(k+1)}{2} + 1 + k = \frac{(k+1)(k+2)}{2}$ disjoint (u, v) -paths in SR_n . Therefore, the theorem is true in this case. Please see figure 4 for illustration.

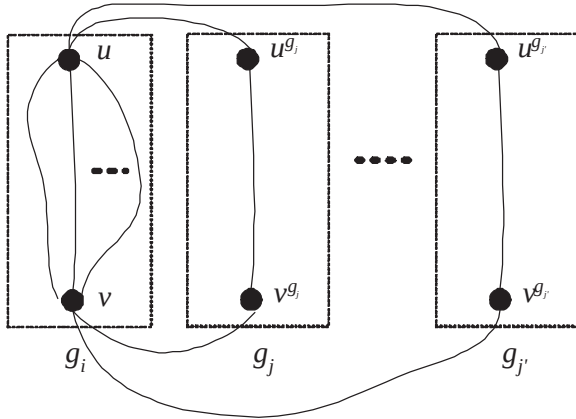


Figure 6: Illustration of Case 1.

Case 2: u and v are in the different group.

Let $u \in V(g_i)$ and $v \in V(g_j)$ with $i \neq j$. Since g_i and g_j are two $[\frac{k(k+1)}{2} + 1]$ -regular maximum-connected graphs, $|V(g_i)| =$

$|V(g_j)| = 2^{k+1} \cdot k! > [\frac{k(k+1)}{2} + 1]$ for $k \geq 1$, and there exists a perfect matching M_2 between $V(g_i)$ and $V(g_j)$, by Theorem 2, $G(g_i, g_j; M_2)$ is $[\frac{k(k+1)}{2} + 2]$ super-connected, i.e., there are $\frac{k(k+1)}{2} + 2$ disjoint paths between u and v of $G(g_i, g_j; M_2)$. By property 5, every group g_l have exactly one vertex u^{g_l} and one vertex v^{g_l} connected to u and v respectively for $g_l \neq g_i, g_j$. Clearly, there exists a path P^{g_l} joining u^{g_l} and v^{g_l} in g_l . Then for $k - 1$ group, there are $k - 1$ disjoint paths joining u and v in $SR_n - (g_i \cup g_j)$. Since $G(g_i, g_j; M_2)$ have $\frac{k(k+1)}{2} + 2$ disjoint (u, v) -paths, there are $\frac{k(k+1)}{2} + 2 + k - 1 = \frac{(k+1)(k+2)}{2}$ disjoint (u, v) -paths in SR_n . Therefore, the theorem is true in this case. Please see figure 4 for illustration. This complete the proof. $\square$

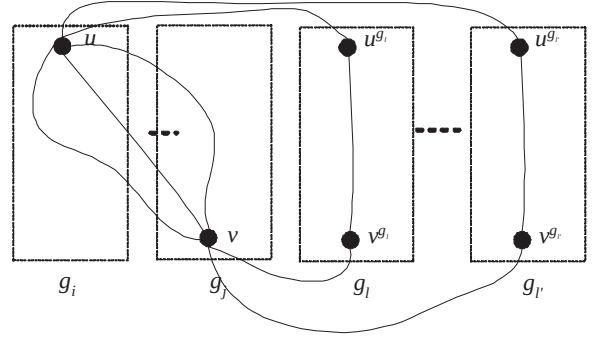


Figure 7: Illustration of Case 2.

Then, we consider the diameter problem of SR_n . By the reversal distance of a signed permutation proposed by Hannenhalli et. al., we can give a bound of the diameter of SR_n as follows.

Theorem 4 $D(SR_n) \leq n + 1$ for $n \geq 1$.

We omit the proofs of the following lemmas to reduce the space.

Lemma 2 Let g_i be a group of SR_n . $D(g_i) \leq D(SR_{n-1}) + 1 \leq n + 1$ for $n \geq 2$.

Lemma 3 Let g_i be a group of SR_n and $k = \frac{n(n-1)}{2}$. $\omega_k(g_i) \leq \omega_k(SR_{n-1}) + 1$.

Lemma 4 Let g_i and g_j be two groups of SR_n and $k = \frac{n(n-1)}{2}$. Let $G = G(g_i, g_j, M)$ be a subgroup of SR_n induced by g_i and g_j . $\omega_k(G) \leq \omega_k(SR_{n-1}) + 1$.

Theorem 5 Let g_i be a group of SR_n . $\omega_k(SR_n) \leq D(SR_n) + 2 \leq n + 3$ with $k = \frac{n(n+1)}{2}$.

Proof. We prove this theorem by induction on n . For $n = 2$, since SR_2 is a 3-dimensional hypercube[10], the k -wide diameter of SR_2 is $3 \leq 2 + 2$. Therefore, the base case is true. Assume that the theorem is true for $n = m$, i.e., $\omega_k(SR_m) \leq m + 2$. Then for $n = m + 1$, we divide the prove into two cases by the positions between any two vertices u and v in SR_n and then use the same proof technique as Theorem 3.

Case 1: u and v are in the same group, say g_i .

By Lemma 3, $\omega_{k-n}(g_i) \leq \omega_{k-(n+1)}(SR_{n-1}) + 1$. Hence, the maximum length of the $\frac{(n-1)(n)}{2} + 1$ disjoint (u, v) -paths is less or equal to $\omega_{k-(n+1)}(SR_{n-1}) + 1 \leq n + 3$ by induction hypothesis. Using the same technique as the case 1 of the proof of Theorem 3, there are $n - 1$ disjoint (u, v) -paths P_j passing the groups g_j for $1 \leq j \neq i \leq n$. By Lemma refdgi, $d_{g_j}(u^{g_j}, v^{g_j}) \leq D(g_j) \leq n + 1$. Hence $\text{len}(P_j) \leq d_{g_j}(u^{g_j}, v^{g_j}) + 2 \leq n + 3$. Therefore, the lengths of all $\frac{n(n+1)}{2}$ (u, v) -paths are less or equal to $n + 3$.

Case 2: u and v are in the different group.

Let $u \in V(g_i)$ and $v \in V(g_j)$ with $i \neq j$ and let $G = G(g_i, g_j, M)$ be a subgroup of SR_n induced by g_i and g_j . By Lemma 4, $\omega_{k-(n-1)}(G) \leq \omega_{k-(n+1)}(SR_{n-1}) + 1$. Hence, the maximum length of the $\frac{(n-1)(n)}{2} + 2$ disjoint (u, v) -paths is less or equal to $\omega_{k-(n+1)}(SR_{n-1}) + 1 \leq n + 3$ by induction hypothesis. Using the same technique as the case 2 of the proof of Theorem 3, there are $n - 2$ disjoint (u, v) -paths P_l passing the groups g_l for $1 \leq l \neq j \neq i \leq n$. By Lemma refdgi, $d_{g_l}(u^{g_l}, v^{g_l}) \leq D(g_l) \leq n + 1$. Hence $\text{len}(P_l) \leq d_{g_l}(u^{g_l}, v^{g_l}) + 2 \leq n + 3$. Therefore, the lengths of all $\frac{n(n+1)}{2}$ (u, v) -paths are less or equal to $n + 3$. This complete the proof. $\square$

Let $F(SR_n)$ be the faulty vertex set in SR_n . Since the connectivity $\kappa(SR_n) = \frac{n(n+1)}{2}$, $|F(SR_n)| \leq \frac{n(n+1)}{2} - 1$. If $|F(SR_n)| \geq \frac{n(n+1)}{2}$, then SR_n may disconnected and two vertices may have no path joining them. Therefore, we discussed the fault diameter under the condition f such that $f : |F(SR_n)| < \frac{n(n+1)}{2}$. Therefore, for the fault diameter of SR_n , we have the following results.

Theorem 6 $D^f(SR_n) \leq \omega_k(SR_n) \leq D(SR_n) + 2 \leq n + 3$ with $k = \frac{n(n+1)}{2}$.

Corollary 1 Let π be a permutation of n -bit numbers. The k -fault reversal distance of a permutation π is $d_k^f(\pi) \leq n + 3$ with $k < \frac{n(n+1)}{2}$.

5 Conclusion

In this paper, we introduce a new graph to represent sorting signed permutation by reversals problem, Signed Reversal Graph, SR_n . We also prove the connectivity and give a bound of diameter of SR_n . Moreover, we also introduce the fault reversal distance problem for sorting by reversals problem and we can using some traditional graph problems, fault distance and fault diameter, to discusse this problem. SR_n has some interesting properties such as symmetric, recursive structure, low diameter and maximal connectivity. In the future, we will continue to try to model other genome rearrangement problems into graph models.

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